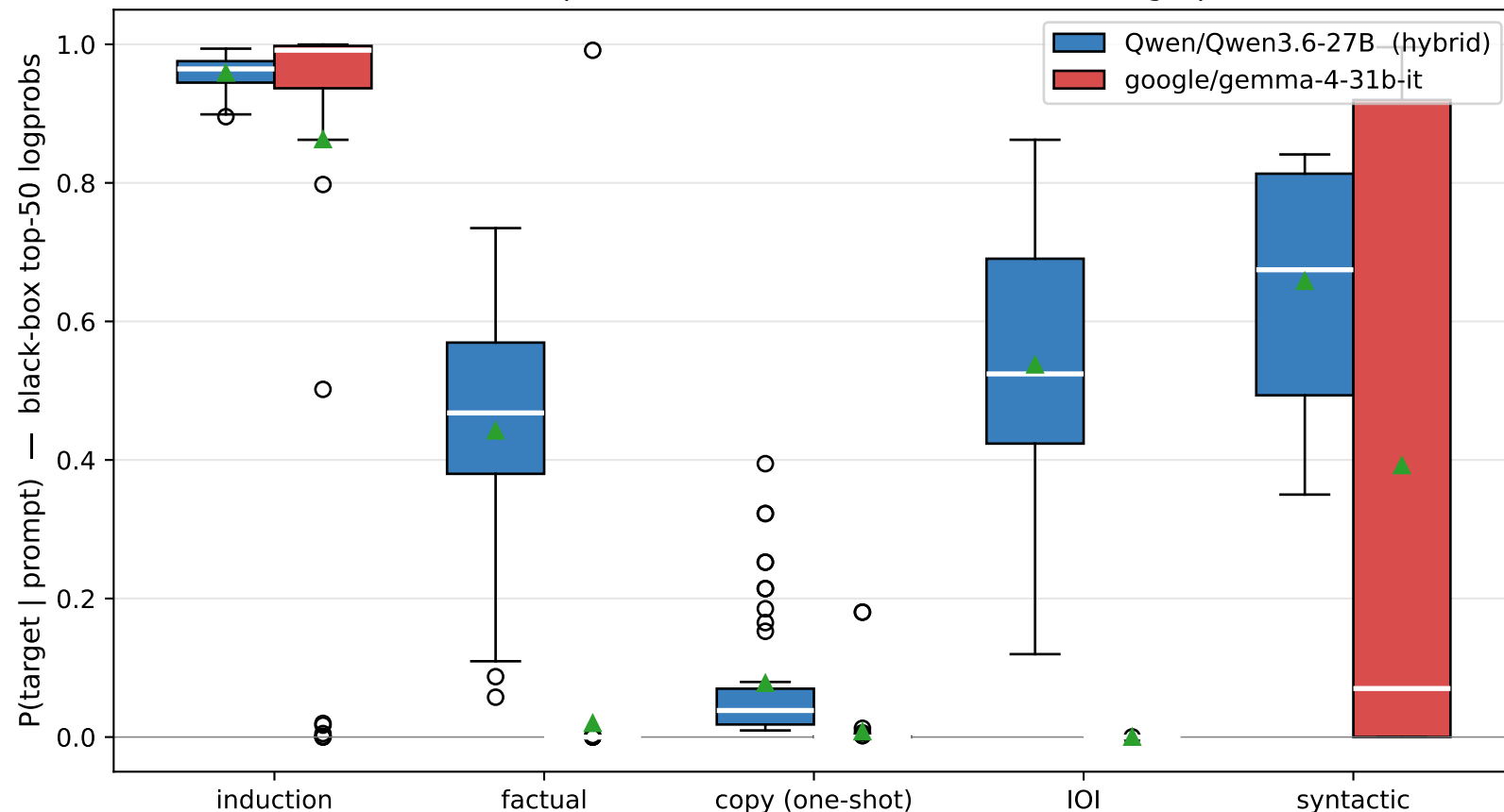


H31 — black-box LLM behavioral interpreter on 2 frontier LLMs 256 probes • 5 circuit classes • 0 hooks • dual-verified z3 + cvc5

H31 — same probes, different vendor behavioral fingerprint



Qwen3.6–27B factual ($R^2 = 0.89$, depth-4 EML)

$$P_{\text{target}} \approx 0.5954 - 0.1353 \cdot \text{eml}(L, \text{eml}(L-H, 1))$$

L = induction lag (here = 0)

H = entropy of top-50 (working: 1.94–2.40)

Cert pair ($\tau=0.10$):

- $P > \tau$ over IQR working box
z3: **unsat** (12 ms)
cvc5: **unsat** (4 ms)
- $P > \tau$ over failure box
z3: **sat** (3 ms) + model
cvc5: **sat** (4 ms) + model

Black–box discipline:

no hooks, no output_attentions,
no output_hidden_states, top-K only